

Varun Gangavarapu

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EDUCATION

Rutgers the State University of New Jersey

New Brunswick, NJ

Bachelor of Science in Computer Science

September 2022 – May 2026

- Design & Analysis of Algorithms, Computer Architecture, Internet Technology, Software Methodology

EXPERIENCE

Amazon

June 2026 – Present

Software Development Engineer

Seattle, WA

- Replaced a **greedy algorithm** in a real estate allocation optimizer with a **Mixed Integer Program** solver using **McCormick Linearization** to reward adjacent placements achieving a **12%** clustering gain at no additional relocation cost
- Rearchitected the allocation optimizer from a single-process solve into an AWS Batch pipeline that fans out **23 concurrent solvers** sized at runtime, cutting wall-clock runtime **5x** and optimizer timeouts from **30% to 4%**.
- Codified cross-service architecture, data contracts and recurring failure patterns into agent context and rule sets spanning **20 frontend and backend modules** across **5 services**, increasing merged PRs per engineer by **30%**.
- Engineered a code review analysis pipeline that mined **15,000+ reviewer comments** into automated rules for a **code verification agent**; backtesting confirmed **500+ defects** at an **84%** accept rate, **46%** of which engineers had missed.

Amazon

May 2025 – August 2025

Software Development Engineering Intern

Bellevue, WA

- Built a **serverless digital twin platform** providing real-time operational visibility across **64 Amazon fulfillment centers** with **99% uptime** and global scalability via **CloudFront**.
- Developed an **authenticated** Cloudscape React frontend with deep-linking, multi-mode rendering, and bidirectional URL state sync, powering **100+ embedded dashboards** across operations and BI via **QuickSight**.
- Designed a **configuration-driven** onboarding system using **S3, SQS, OSIS**, and **OpenSearch** with **YAML** configs and **DynamoDB** DSL templates; ingests **5M+ rows/day** across **25 metrics** with zero code changes.
- Implemented a **high-performance** backend using **API Gateway, Lambda, OpenSearch**, and **DynamoDB** query templating; achieved **sub-500ms** response times with cross-AZ fault tolerance and idempotent ingestion.
- Contributed to a **patent** for automated digital-twin generation by reverse-engineering proprietary contractor layout logic, unblocking **3 fulfillment centers** and avoiding **\$3M+** in manual drafting costs.

PROJECTS

Dynamic Vector Based Spotify Recommendation Engine

March 2024

Python, JavaScript, Flask, OAuth

[!\[\]\(9f3852d68d41e1e95bc4ec10e81aba4b_img.jpg\) GitHub](#)

- Secured **1st place** in a hackathon with **500 entrants**, leading full-stack development of a **Vector-Based Spotify Recommendation Engine**.
- Conducted advanced **sentiment analysis** by integrating linguistic data from Spotify and MusixMatch APIs with a **HuggingFace transformer** to generate a “**Sentiment Vector**” in a **24-dimension** space.
- Modeled a **vector database** in **PineCone** utilizing **KNN algorithms** for efficient Sentiment Vector retrieval, enhancing song recommendation accuracy and performance in a user-centric application.
- Independently built Front End with **JavaScript, HTML**, and **CSS** incorporating advanced animations with the use of the **GSAP** library along with **Swiper CSS** and **Locomotive Scroll**.

Course Reveal

January 2024

React/Next.js, TailWind CSS, Python, Flask

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- Deployed a full-stack platform where students upload course schedules to find classmates in shared sections, reaching **100s of students** across registration cycles.
- Implemented the matching engine that normalizes uploaded schedules into course sections and resolves shared enrollments through inverted-index lookups over a **5,000+ course** catalog.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, Go, C++, SQL, HTML, CSS

Frameworks/Libraries: React, Next.js, Node.js, Spring Boot, Flask, Express, Tailwind CSS, Pandas, NumPy, PyTorch

AI/Agents: Agent Tooling & Rule Sets, MCP, RAG, Vector Databases (Pinecone, OpenSearch), Embeddings

Cloud/Infrastructure: AWS (Lambda, API Gateway, DynamoDB, OpenSearch), Docker, Kubernetes, Git, Linux